

BM17-D262 - The Box

A low-cost electromagnetic-reduction chamber and QC protocol for exploratory PR-AYC-G psychophysiology

Base Model 1.7 infrastructure layer | v1.0

Document status: Research infrastructure draft. This is not a medical device, not a clinical protocol, not electrical-safety certification, not a validated RF-shielded-room specification, and not proof of any PR-AYC-G endpoint.

1. Purpose

This document formalizes the uploaded '10x10x8 Stasis Chamber: Master Build Plan & Testing Protocol' as a Base Model 1.7 infrastructure layer. The source describes a 10x10x8 Faraday-style tent for high-resolution, dual-subject EEG hyperscanning with OpenBCI Mark IV; it lists PVC framing, copper mesh, conductive copper tape, an 8-foot copper grounding rod, and dielectric floor isolation as core build elements.

The Box is formalized here as an environmental-control layer: a build, logging, and QC architecture intended to reduce avoidable electromagnetic and procedural noise before EEG, HRV, respiration, stimulus, and self-report data are interpreted.

Core rule: The Box is not evidence. The Box is the stage on which evidence is asked to appear.

2. Position inside Base Model 1.7

Layer	Role in Base Model 1.7	Permitted claim
Environmental-control layer	Reduces room, cable, power, and RF/line-noise uncertainty before analysis.	May improve pilot data quality if QC confirms noise reduction.
Measurement-stage infrastructure	Documents chamber geometry, grounding, stimulus delivery, cable routing, and acquisition conditions.	Makes the room part of the method rather than an invisible confound.
PR-AYC-G acquisition support	Supports OpenBCI/LSL/LabRecorder workflows by creating a documented low-noise recording environment.	Does not replace stream validation, marker validation, or artifact rejection.
Citizen-neuroscience scaffold	Provides a reproducible build vocabulary for low-cost exploratory EEG settings.	Democratizes method literacy, not clinical authority or proof of consciousness.

3. Source-derived build components

Component class	Source-derived description	Base Model handling
Frame	1.5-inch Schedule 40 PVC pipe and furniture-grade fittings for the 10x10x8 skeletal chamber.	Log final dimensions, joints, bracing, and photos before use.
Conductive envelope	Woven copper mesh, preferably wide rolls, with conductive copper foil tape for seams.	Treat seam continuity as a QC object. Do not infer shielding without tests.
Grounding	Exterior 8-foot copper grounding rod, brass clamp, and lower perimeter grounding bond.	Requires electrical-safety review. Record ground architecture and do not connect participants to mains-powered systems.
Dielectric floor layer	Concrete curing blankets or equivalent insulating barrier beneath copper floor mesh.	Treat as mechanical/electrical isolation from concrete/rebar and document installation.
Door/flap	Oversized mesh flap with magnetic hem sealing sides and bottom.	Door seams are high-risk leak points and require repeated QC.

4. Build phases

1. Ground and foundation: establish the ground plan, dielectric floor barrier, distance from metal walls, and all safety exclusions before copper or electronics are added.
2. Skeletal frame: assemble and document the bottom square, verticals, top square, and roof cross-bracing.
3. Conductive envelope: install wide copper mesh panels with fold-crush-tape seams and bond wall panels to the floor mesh.
4. Door and perimeter ground: build the oversized flap, magnetic or equivalent closing seam, lower perimeter bond, and exterior ground clamp.
5. Proof-of-concept testing: perform AM/radio, cellphone, and OpenBCI 60 Hz checks before human pilot work.

5. Minimum QC suite

QC test	Purpose	Pass/fail framing
AM radio static test	Crude low-frequency leak check using a battery-powered radio tuned to static.	Pilot screen only. Silence/reduction is supportive but not quantitative attenuation.
Cellphone drop test	Crude high-frequency RF ingress check using a cellular call attempt.	Call failure is supportive. Phone ringing indicates RF penetration or seam weakness.
OpenBCI 60 Hz FFT test	Directly relevant EEG noise-floor check with the cap/headset inside the sealed chamber.	60 Hz line should be reduced relative to outside/baseline. Quantify with PSD ratio when possible.
Door-open/door-closed PSD	Tests whether closure measurably changes line noise.	If no change, the enclosure may not be functioning for relevant bands.
Monitor on/off PSD	Tests stimulus hardware contamination.	If monitor adds line/gamma artifacts, reposition/power/filter before human use.
XDF stream integrity smoke test	Ensures EEG, markers, HRV,	Any zero-sample critical stream

	respiration are all recorded.	invalidates the physiology run.
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6. Quantified environmental metrics

The qualitative tests should be supplemented by measurable PSD and stream-integrity metrics wherever possible.

Metric	Formula / operational definition	Interpretation
Line-noise attenuation	$A_{60} = 10 \log_{10}(P60_{\text{outside}} / P60_{\text{inside}})$	Positive values indicate lower 60 Hz power inside the sealed chamber.
Door closure effect	$D_{60} = 10 \log_{10}(P60_{\text{door_open}} / P60_{\text{door_closed}})$	Positive values indicate that sealing the door reduces line noise.
Stimulus-hardware effect	$S_{60} = 10 \log_{10}(P60_{\text{monitor_on}} / P60_{\text{monitor_off}})$	Positive values indicate monitor/power contamination risk.
Sample-count validity	$N_{\text{samples}} > 0$ for EEG, markers, HRV, respiration where required	Zero EEG samples invalidate EEG analysis even if stream header is visible.
Effective sampling rate	$N_{\text{samples}} / \text{duration_seconds}$	Should be near the expected acquisition rate after accounting for drift and gaps.

7. Safety boundaries

- Have grounding, cable routing, and any connection to earth or building electrical systems reviewed by a qualified electrician or other competent electrical-safety professional.
- Use battery-powered systems whenever possible. Do not place participants in contact with mains-powered equipment or exposed conductors.
- Maintain ventilation, emergency egress, lighting, trip-hazard control, fire safety, and a rapid stop/exit procedure.
- Do not run human-subject experiments without consent, risk disclosure, and appropriate ethics/IRB review where applicable.
- Do not describe the chamber as clinically validated, TEMPEST-certified, medically effective, or diagnostically meaningful.
- Treat copper seams, ground connections, and cable penetrations as failure points requiring repeated inspection.

8. Falsifiers and failure modes

- The 60 Hz line is not reduced inside the chamber relative to outside or door-open baselines.
- The cellphone/radio tests fail and no seam or door correction improves them.
- Stimulus hardware creates a larger artifact than the chamber removes.
- EEG stream headers appear but critical streams record zero samples.
- Participant movement, heat, discomfort, or psychological demand introduced by the chamber overwhelms any noise-floor gain.
- The chamber changes the social/performative state more than it improves measurement quality.

9. Required reporting language

Recommended: "Recordings were conducted inside a documented Faraday-style electromagnetic-reduction chamber. The chamber is treated as a pilot environmental-control apparatus, not as a certified shielded room. Attenuation and stream-integrity checks are reported separately."

Avoid: "The chamber proves isolation," "the room eliminates all artifact," or "the chamber validates the theory."